WAGER: Within-cell Antisymmetric Gain Evaluation of Reasoning

Quang-Vinh Dang[1]

[1]British University Vietnam, Hung Yen, Vietnam.

Contributing authors: vinh.dq4@buv.edu.vn;

Abstract

Vision benchmarks with strong annotation priors can reward a new model even when its gain does not come from better instance-specific visual prediction. Existing audits usually compare a model with a frequency baseline, which confounds two questions: how well each system fits the prior, and whether the *difference* between systems uses information beyond it. We introduce WAGER, **W**ithin-cell **A**ntisymmetric **G**ain **E**valuation of **R**easoning, a new pairwise attribution algorithm. For two frozen probabilistic models, WAGER forms their proper-score contrast, matches examples that share a declared prior feature, and crosses their labels. This yields an exact finite-sample identity: observed model gain equals a prior-transported gain plus an antisymmetric instance-alignment gain. The latter is an order-two U-statistic and, with the quadratic score, equals twice the within-prior-cell prediction–label covariance gained by the new model, connecting it to the classical resolution term of proper-score decomposition. It is computed in $\boldsymbol{O(NK)}$ time, requires no fitted prior model or projection fold, and admits image-cluster-robust inference and a cell-stratified randomization test. We prove two further exact results: the leave-one-out transport removes a finite-sample attenuation bias of $\boldsymbol{(n_c - 1)/n_c}$ that a naive in-sample plug-in prior would incur, and coarsening the audit cell changes the estimate by a between-cell covariance term identified from first principles. Controlled experiments recover injected instance signal monotonically and obtain type-I error **0.030** at level **0.05**. On 229,605 Visual Genome PredCls relations, a class-only MLP improves the frequency baseline entirely through the prior-transported channel, whereas adding box geometry produces an instance-alignment gain of **0.01439** (95% CI $\boldsymbol{[0.01373, 0.01504]}$; randomization $\boldsymbol{p =}$ **0.002**). Applying the same estimator unchanged to long-tailed image classification on CIFAR-100-LT separates two re-weighting schedules that aggregate scores cannot: class-balanced re-weighting from initialization yields a near-zero total gain ($\boldsymbol{+0.00143}$) that conceals a $\boldsymbol{+0.19713}$ prior gain cancelling a $\boldsymbol{-0.19569}$ alignment loss, while the deferred schedule's genuine $\boldsymbol{+0.06606}$

improvement is roughly two-thirds prior-recoverable and one-third instance alignment (**+0.02400**, 95% CI [**0.01340, 0.03460**]), concentrated on rare classes. WAGER therefore attributes a model-to-model improvement directly, without treating a separately estimated baseline as ground truth.

Keywords: Visual Reasoning, Dataset Bias, Counterfactual Evaluation, U-statistics, Randomization Tests, Scene Graph Generation

1 Introduction

Computer-vision benchmarks often contain a shortcut that is both powerful and easy to state. In scene graph generation (SGG), the ordered object-class pair strongly predicts the relation label; in visual question answering (VQA), question form predicts the answer; in long-tail recognition, category frequency dominates aggregate performance. Models can therefore improve a headline score by fitting annotation regularities more precisely without improving the instance-specific association between an image and its label. This is not merely hypothetical: frequency baselines are competitive on Visual Genome [1], VQA systems fail under changed answer priors [2, 3], standard metrics obscure tail behavior [4], and the same diagnosis continues to reappear in current work: open-vocabulary and knowledge-enhanced SGG methods still report a single aggregate score [5–7], long-tailed recognition methods rebalance training toward rare classes without attributing a resulting gain to the frequency prior versus instance-specific evidence [8–10], and recent benchmark audits show multimodal models can exploit non-visual shortcuts that inflate exactly this kind of aggregate score [11, 12].

The usual audit trains or constructs a prior-only baseline and compares its score with a full model. That strategy is useful diagnostically, but it does not identify where the *gain between two submitted systems* came from. Baseline estimation error, smoothing, calibration, and model-family mismatch all enter the result. More fundamentally, separately decomposing each model and subtracting the decompositions is an indirect answer to a pairwise question.

We instead ask a counterfactual question at the level of the gain itself. Let f_0 be the old model, f_1 the claimed improvement, and $\phi(x)$ the declared prior feature. For two test examples i, j with the same ϕ, we retain both models' prediction vectors but cross the observed labels. The crossed assignment preserves the prior cell and its label histogram while destroying which prediction vector belongs with which label. Comparing the observed and crossed assignments measures whether $f_1 - f_0$ improves this instance-level alignment. Figure 1 summarizes the idea.

This construction gives WAGER: *Within-cell Antisymmetric Gain Evaluation of Reasoning*. "Reasoning" is used operationally: predictive alignment with the correct instance beyond ϕ, not a claim of human-like cognition or causality. WAGER makes five contributions.

1. We define a direct model-pair estimand whose prior component is generated by within-cell label transport, not by fitting a frequency forecaster.

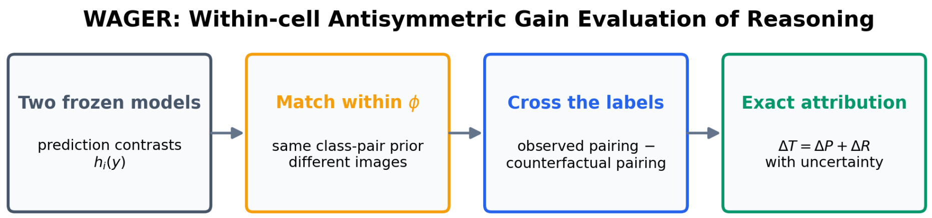

Fig. 1 WAGER audits the gain directly. Prediction contrasts from two frozen models are matched within a declared prior cell, evaluated under observed and crossed label assignments, and separated exactly into prior-transported and instance-alignment components.

2. We prove an exact population and finite-sample decomposition. The residual is an antisymmetric U-statistic; under the quadratic score it has a covariance identity that identifies it with the resolution term of the classical proper-score decomposition [13–15], applied to a model-pair contrast rather than to one forecaster.
3. We prove two further exact results that were not available for the single-model decomposition: leave-one-out transport removes a finite-sample attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in prior would otherwise incur at every finite cell size (Proposition 2), and coarsening the audit cell changes the estimate by an explicit between-cell covariance term (Proposition 1), which turns an empirical sensitivity finding into a population-level mechanism.
4. We derive an $O(NK)$ algorithm, a Hájek influence function for image-cluster-robust intervals, and a cell-stratified randomization test under a sharp exchangeability null.
5. We validate the estimator under controlled alternatives and on Visual Genome, where it distinguishes a class-only gain from gains created by adding spatial evidence, and further from gains created by a model that consumes real image pixels through a frozen CLIP encoder rather than only box geometry (Section 4.5).
6. We show WAGER is not SGG-specific by applying it unchanged to a second domain and task type – long-tailed image classification on CIFAR-100-LT – decomposing the gain of a class-balanced classifier over a vanilla cross-entropy baseline into class-frequency-prior and instance-alignment components (Section 4.6), giving empirical support to the broader-applicability claim of Section 5.2 rather than leaving it as an untested hypothesis.

The redesign is important in what it removes. WAGER does not estimate a model's "self-prior," does not require a projection/evaluation split, does not equate betting growth with information, and does not rely on an arbitrary threshold for a gain ratio. The primary quantity is the signed instance-alignment gain with uncertainty.

2 Related Work

2.1 Benchmark priors and shortcut learning

Dataset bias has long complicated comparisons between visual recognition systems [16, 17]. In SGG, the FREQ predictor estimates $p(\text{predicate} \mid \text{subject}, \text{object})$ and is surprisingly competitive [1]; mean recall and causal interventions were proposed to reduce this dependence [18–20], although metric pathologies remain [4]. More recent SGG work continues this line under open-vocabulary and knowledge-enhanced settings [5–7], and still reports gains as a single aggregate score rather than attributing them to prior fitting versus instance-specific evidence. VQA-CP changes language priors between train and test [3], but changing-prior splits can themselves be exploited [21]; recent VQA work continues to target the same language-prior shortcut through contrastive and relation-aware objectives [22, 23]. Benchmark-level audits of exploitable non-visual shortcuts have also expanded beyond VQA to broader multimodal evaluation [11, 12], reinforcing the same diagnosis WAGER targets: an aggregate score can rise for reasons unrelated to instance-specific visual evidence. These methods alter training, data, or metrics, or audit a single model's exploitability against a benchmark's own structure. WAGER instead audits a fixed pair of probabilistic predictors on a fixed test set, attributing their score difference rather than characterizing either model alone.

2.2 Long-tailed recognition

A structurally identical shortcut appears outside SGG: in long-tailed visual recognition, a classifier can improve aggregate accuracy by fitting the training-time class-frequency prior more precisely, without improving instance-specific discrimination on any individual class [8, 24]. Class-balanced re-weighting, logit adjustment, and feature-fusion methods address this by rebalancing the training objective or representation toward rare classes [8–10], analogous to how mean-recall and causal-intervention methods rebalance SGG training. As in SGG, these methods change how a model is trained; they do not attribute a fixed pair of trained models' accuracy gap to frequency-fitting versus instance-specific improvement. Section 4.6 applies WAGER unchanged to this setting, taking the benchmark's own coarse superclasses as the declared prior feature ϕ, and separates two re-weighting schedules that aggregate scores cannot tell apart: one whose near-zero total gain conceals a large prior improvement cancelling an equally large alignment loss, and one whose genuine improvement is part prior-refitting and part rare-class alignment.

2.3 Information and predictive probing

Usable-information analyses measure information accessible to a chosen predictor family [25–27]. Their central objects are model- or family-level quantities. WAGER is deliberately narrower and more operational: it attributes the proper-score difference between two concrete frozen models. We do not identify conditional mutual information and do not claim that a positive residual establishes causal or compositional reasoning.

2.4 Proper scores, randomization, and U-statistics

Strictly proper scoring rules reward honest predictive distributions [28]; WAGER uses their model-pair contrast. A recent survey revisits estimation and forecast evaluation under proper scoring rules broadly [29], and the reliability–resolution–uncertainty partition itself continues to be re-derived and extended: to general strictly proper scores and multiclass outcomes [30], to entropy/divergence-based uncertainty quantification [31], and via an alternative covariance rearrangement for the Brier score specifically [32]. WAGER uses the same partition but applies it to a paired gain contrast rather than to one forecaster, as detailed in Section 2.5. Permutation methods have been used to assess classifier performance and conditional independence [33, 34], with recent work characterizing when generalized permutation tests attain optimal power [35] and extending conditional-independence testing via ball-divergence measures [36] and transport-map reductions to the unconditional case [37]. Conditional permutation tests typically estimate or assume a conditional distribution when confounders are continuous or high-dimensional. WAGER exploits a different setting: the benchmark supplies a discrete audit cell ϕ, and the target is not generic conditional independence but a specific model-to-model score gain. This permits exact within-cell transport without estimating a nuisance law. The antisymmetric residual is an order-two U-statistic in the sense of Hoeffding [38]; its Hájek-projection inference is part of the same broader program of finite- and large-sample U-statistic theory that recent work extends to anytime-valid confidence sequences [39], enabling an efficient all-pairs estimator and influence-function inference here. Pairwise betting has also been used to test exchangeability sequentially [40]; WAGER is a fixed-sample gain decomposition and makes no sequential-validity claim. Its new element is the gain-contrast transport identity and the resulting attribution algorithm, rather than permutation or U-statistic theory by itself.

2.5 Relation to classical proper-score decompositions

The covariance identity in Eq. (7) is a relative of the reliability resolution–uncertainty partition of a single proper score [13, 14], later generalized and made conditioning-set explicit by Bröcker [15]: conditioned on a chosen grouping variable, a proper score splits into a calibration term and a resolution term, and the resolution term is a covariance between the forecast and the outcome indicator within groups. We deliberately build on this fact rather than obscure it. WAGER is not a new proper score, and ΔR_c is best understood as a *resolution difference*, not a new quantity to be evaluated on its own. Two things distinguish WAGER from applying the classical decomposition to q_0 and q_1 separately and subtracting the two resolutions. First, that indirect route requires two resolution estimates, each of which is generally biased in finite samples by the same self-prediction mechanism formalized in Section 3.6; the bias does not cancel under subtraction because it scales with each model's own cell counts. WAGER instead forms the paired gain vector $H_x(y)$ first and decomposes the *contrast* in one pass, which is what makes the exact finite-sample identity and the antisymmetric U-statistic kernel possible. Second, the classical decomposition is a diagnostic for a single forecaster's calibration and sharpness; it neither targets nor equips a specific pairwise comparison with inference. WAGER supplies image-cluster-robust confidence intervals and a cell-stratified randomization test for exactly the gain a benchmark submission claims. The technical contribution here is narrower than a new form of score decomposition: it is the transport construction and finite-sample identity that convert a known resolution concept into a model-to-model attribution algorithm with an exactness and inference guarantee that the single-model decomposition does not provide.

3 Method

3.1 Setup: audit a gain, not a model

Let $q_0(\cdot \mid x)$ and $q_1(\cdot \mid x)$ be the predictive distributions of an old and a new frozen model over K labels. Let $S(q, y)$ be a bounded proper score, oriented so that larger is better, and let $C = \phi(X)$ be a declared discrete prior feature. For SGG, C is the ordered subject–object class pair. Define the instance-specific *gain vector*

$$H_x(y) = S(q_1(\cdot \mid x), y) - S(q_0(\cdot \mid x), y), \qquad y = 1, \dots, K. \tag{1}$$

Unlike the realized gain $H_X(Y)$, this vector can be evaluated at a counterfactual label without rerunning either model.

Our default is the quadratic score

$$S_{\mathrm{B}}(q, y) = 2q_y - \|q\|_2^2, \tag{2}$$

which differs from negative Brier loss only by a label-only constant. It is strictly proper and bounded, avoids probability clipping, and uses the entire predictive distribution. A floored log score is a sensitivity analysis.

3.2 Within-cell counterfactual transport

For a prior cell c, define three population quantities:

$$\Delta T_c = \mathbb{E}[H_X(Y) \mid C = c], \tag{3}$$
$$\Delta P_c = \mathbb{E}_{H|c}\, \mathbb{E}_{Y'|c}[H_X(Y')], \tag{4}$$
$$\Delta R_c = \Delta T_c - \Delta P_c. \tag{5}$$

Here Y' is an independent label drawn from the same cell. Thus ΔT_c is the observed score gain, ΔP_c is the gain retained after breaking the instance-specific prediction–label assignment while preserving the prior cell, and ΔR_c is the alignment gain lost by that transport.

Theorem 1 (Transport and covariance identities)**.** *For any integrable score contrast H,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \qquad \Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)\,. \tag{6}$$

For the quadratic score, writing $\Delta q_y(X) = q_1(y \mid X) - q_0(y \mid X)$,

$$\Delta R_c = 2 \sum_{y=1}^{K} \mathrm{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} \mid C = c)\,. \tag{7}$$

Consequently, if the gain vector is a function of c *alone, then* $\Delta R_c = 0$.

Equation (7) is the operational meaning of WAGER: the new model receives reasoning credit only to the extent that its probability change aligns more strongly with the label of the correct instance within the same prior cell. Cell-constant score improvements, including improved fitting of the cell's label histogram, remain in ΔP. The result does not assert that every nonzero covariance is causal reasoning; unrecorded within-cell shortcuts can also contribute, so ϕ must be declared and stress-tested.

3.3 Coarsening the audit cell

Section 4.4 reports that replacing the class-pair cell with a coarser subject-only cell changes the estimated alignment gain. The following result gives that sensitivity an exact population-level mechanism rather than treating it as an empirical curiosity.

Proposition 1 (Coarsening decomposition)**.** *Let* $\bar{\phi} = g(\phi)$ *be any coarsening of the audit feature, so that each cell* $\bar{c}$ *of* $\bar{\phi}$ *is a union of cells of* ϕ*. Then*

$$\Delta R_{\bar{c}} = \mathbb{E}\big[\Delta R_C \mid \bar{\phi} = \bar{c}\big] + \sum_{y=1}^{K} \mathrm{Cov}\big(\mathbb{E}[H_X(y) \mid C],\, \Pr(Y = y \mid C) \bigm| \bar{\phi} = \bar{c}\big)\,. \tag{8}$$

The first term is the coarse cell's share of the fine-grained alignment gain the paper already reports; it is the quantity a reader would expect a coarser audit to recover on average. The second term is new: it is a between-cell covariance, taken over the finer cells nested inside $\bar{c}$, between each fine cell's typical gain at label y and its label-y frequency. It is generally not sign definite, so coarsening ϕ can inflate or deflate the credited alignment gain relative to the finer audit, and Table 3 shows the empirically inflating case. Because this second term vanishes identically once ϕ is refined to make it constant on every non-vanishing fine cell (in particular, in the limit where ϕ is the finest cell that remains non-singleton for every example), Proposition 1 formalizes why the finest defensible audit feature is the conservative default: coarsening can only add cross-cell prior structure into ΔR, never remove genuine within-cell alignment that the fine partition already isolates. The proof is a per-label application of the law of total covariance, summed over y (Appendix A).

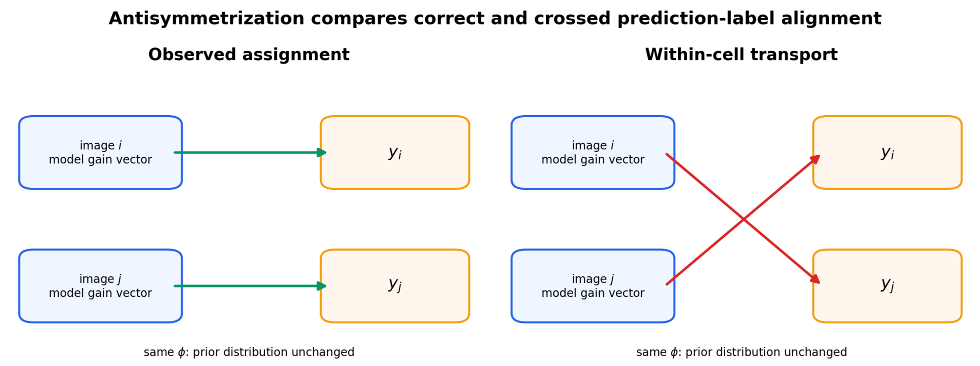

Fig. 2 The antisymmetric pair kernel. For two examples in the same prior cell, WAGER subtracts the score under crossed labels from the score under the observed assignment. Cell membership and label counts are unchanged; only instance alignment is removed.

3.4 Exact finite-sample decomposition

Consider cell c with $n_c \geq 2$ examples. Write $H_i(y) = H_{x_i}(y)$. WAGER computes

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_{i \in c} H_i(y_i), \tag{9}$$

$$\widehat{\Delta P}_c = \frac{1}{n_c(n_c - 1)} \sum_{i \neq j \in c} H_i(y_j), \tag{10}$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j \in c} A_{ij}, \tag{11}$$

where the antisymmetric kernel is

$$A_{ij} = \frac{1}{2}\{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \tag{12}$$

Theorem 2 (Finite-sample WAGER identity)**.** *For every cell with $n_c \geq 2$, without a distributional assumption,*

$$\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c. \tag{13}$$

Moreover, under exchangeable sampling within the cell, $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic for ΔR_c.

The dataset-level quantities weight cells by their number of identified examples: $\widehat{\Delta Z} = N_*^{-1} \sum_{c:n_c \geq 2} n_c \widehat{\Delta Z}_c$ for $Z \in \{T, P, R\}$. Singleton cells cannot identify a within-cell counterfactual; WAGER excludes and reports them instead of inventing a smoothed estimate. If $\widehat{\Delta T} > 0$, the descriptive alignment share is $\widehat{\rho} = \widehat{\Delta R}/\widehat{\Delta T}$. It may exceed one when instance alignment improves while prior fitting deteriorates, and it is undefined for a nonpositive total gain. We therefore report $\widehat{\Delta R}$ as the primary result, not a thresholded ratio.

3.5 Linear-time computation

Naively evaluating Eq. (10) is quadratic in cell size. Let m_{cy} be the count of label y in cell c. For example $i \in c$, its transported score is

$$P_i = \frac{\sum_{y=1}^{K} m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}. \tag{14}$$

Averaging P_i yields Eq. (10); averaging $H_i(y_i) - P_i$ yields Eq. (11). Thus the entire estimator costs $O(NK)$ time and $O(NK)$ storage for cached prediction vectors. No model fitting, projection fold, hierarchy, or smoothing hyperparameter appears.

3.6 Why leave-one-out transport, not a fitted-in-sample prior

Equation (14) excludes example i's own label from its transported score. It is natural to ask what is lost by the simpler alternative of fitting the cell's empirical label frequency on all n_c examples, including i, and using that fitted frequency as the counterfactual, i.e. treating $\hat{p}_c(y) = m_{cy}/n_c$ as a prior model and setting $\tilde{P}_i = \sum_y \hat{p}_c(y) H_i(y) = n_c^{-1} \sum_y m_{cy} H_i(y)$. This is the natural plug-in estimator, and it is what a frequency-collapsed projection baseline computes when no separate evaluation fold is withheld from fitting.

Proposition 2 (Exact attenuation of the in-sample plug-in)**.** *For every cell with $n_c \geq 2$ and every $i \in c$, writing $\hat{P}_i$ for Eq. (14) and $\hat{R}_i = H_i(y_i) - \hat{P}_i$,*

$$\tilde{P}_i = \hat{P}_i + \frac{1}{n_c}\hat{R}_i, \qquad \tilde{R}_i := H_i(y_i) - \tilde{P}_i = \frac{n_c - 1}{n_c}\hat{R}_i. \tag{15}$$

Consequently the in-sample plug-in's dataset-level statistic obeys $\widetilde{\Delta R} = N_*^{-1} \sum_{c:n_c \geq 2} (n_c - 1)\widehat{\Delta R}_c$, a multiplicative shrinkage of every cell's leave-one-out alignment gain by the factor $(n_c - 1)/n_c$.

The shrinkage in Eq. (15) is deterministic, not a variance statement: whenever $\widehat{\Delta R}_c \neq 0$, the in-sample plug-in is biased for it at every finite n_c, with the largest relative attenuation exactly in the smallest identified cells (50% at $n_c = 2$, vanishing only as $n_c \to \infty$). This is why the redesign removes the projection/evaluation split that the rejected precursor of WAGER used to control the same self-prediction bias: a held-out fold restores unbiasedness by fitting $\hat{p}_c$ on data disjoint from the scored examples, but it does so by discarding examples from both stages and by introducing a split-fraction hyperparameter and, typically, higher variance. WAGER's leave-one-out U-statistic removes the bias exactly, at the full sample, for every ϕ-cell with $n_c \geq 2$, without a fitted nuisance distribution at all. Section 2.5 situates this result against the

Algorithm 1 WAGER gain decomposition

Require: frozen q_1, q_0; labels y; prior cells ϕ; proper score S
Ensure: $\widehat{\Delta T}, \widehat{\Delta P}, \widehat{\Delta R}$ and uncertainty

$H_i(k) \leftarrow S(q_{1i}, k) - S(q_{0i}, k)$ for all i, k
for each cell c with $n_c \geq 2$ **do**
  count labels $m_{c1}, \ldots, m_{cK}$
  for each $i \in c$ **do**
    $P_i \leftarrow (\sum_k m_{ck} H_i(k) - H_i(y_i))/(n_c - 1)$
    $R_i \leftarrow H_i(y_i) - P_i$
  end for
end for
$\widehat{\Delta T} \leftarrow \text{mean}\, H_i(y_i)$; $\widehat{\Delta P} \leftarrow \text{mean}\, P_i$; $\widehat{\Delta R} \leftarrow \text{mean}\, R_i$
compute image-clustered Hájek interval and cell-shift randomization p

classical single-model score decomposition, where the analogous self-prediction bias affects each model's resolution estimate separately.

3.7 Inference

For point estimation we use all unordered within-cell pairs. For uncertainty, define $\bar{A}_i = (n_c - 1)^{-1} \sum_{j \neq i} A_{ij}$ and $\widehat{\Delta R}_c = n_c^{-1} \sum_i \bar{A}_i$. The estimated Hájek influence contribution for random cell membership is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \tag{16}$$

We sum $\widehat{\psi}_i$ within image and use the resulting one-way sandwich variance, so multiple relations from the same image are not treated as independent. Total-gain influence is $H_i(y_i) - \widehat{\Delta T}$; prior-gain and share intervals follow from the exact identity and delta method.

As a complementary finite-group diagnostic, WAGER applies an independently sampled cyclic shift to labels inside every cell. Each shift preserves cell size and label histogram. Recomputing $\widehat{\Delta R}$ under B shifts gives

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{\widehat{\Delta R}^{(b)} \geq \widehat{\Delta R}\}}{B + 1}. \tag{17}$$

This is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because SGG relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval; the randomization test is reported as a secondary relation-level diagnostic.

4 Experiments

We evaluate five questions: (i) does WAGER remain centered under the null and respond monotonically to known instance signal; (ii) does it distinguish class-prior

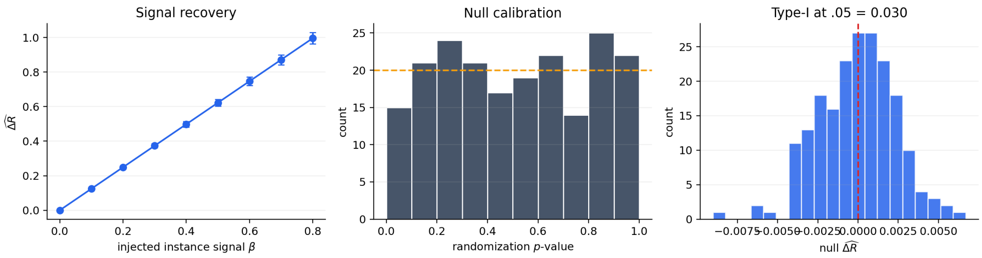

Fig. 3 Controlled validation. WAGER recovers injected within-cell signal monotonically (mean $\pm$ one standard deviation), produces approximately uniform null randomization p-values, and controls type-I error in 200 null runs.

fitting from spatial evidence on a real long-tailed benchmark; (iii) are conclusions stable to score, cell granularity, and sparse-cell filtering; (iv) does the alignment channel behave sensibly when a model consumes real image pixels rather than annotation-derived geometry; and (v) does the same estimator, unchanged, produce a meaningful attribution on a structurally different task in another domain? Questions (i)–(iii) use Visual Genome and controlled simulations; (iv) adds a frozen CLIP encoder on real VG images; (v) moves to long-tailed image classification on CIFAR-100-LT. All confidence intervals are 95%.

4.1 Controlled validation

We generate 24 prior cells with five labels and draw each cell's label distribution from a Dirichlet law. The old model is exactly the cell prior. The new model is $q_1 = (1-\beta)q_0 + \beta q_{\text{oracle}}$, where q_{oracle} assigns 0.98 to the realized label. Forty repetitions are run at each $\beta \in \{0, .1, \ldots, .8\}$. The estimated alignment gain grows linearly and monotonically with the injected signal (Figure 3, left).

For type-I evaluation, we add mean-zero perturbations to prior predictions and randomly reassign the perturbed prediction vectors within each cell, ensuring no systematic prediction–label alignment. Across 200 datasets, the 99-shift randomization test rejects in 0.030 of runs at level 0.05; the mean estimated alignment gain is -8.97×10^{-5}. The null p-values are approximately uniform and null estimates remain centered at zero (Figure 3, center and right).

4.2 Visual Genome setup

We use Visual Genome [41] in the standard VG150 PredCls regime: ground-truth boxes and object labels are supplied, and the model predicts one of 50 predicate labels for each ordered object pair. The training/test split contains 532,987/229,605 relations from 65,228/27,955 images. The prior cell is $\phi = (\text{subject class}, \text{object class})$; Figure 4 shows two real examples from one such cell. Of 8,614 test cells, 6,346 contain at least two relations. WAGER identifies 227,337 test relations (99.0%); the 2,268 singleton cells are reported and excluded.

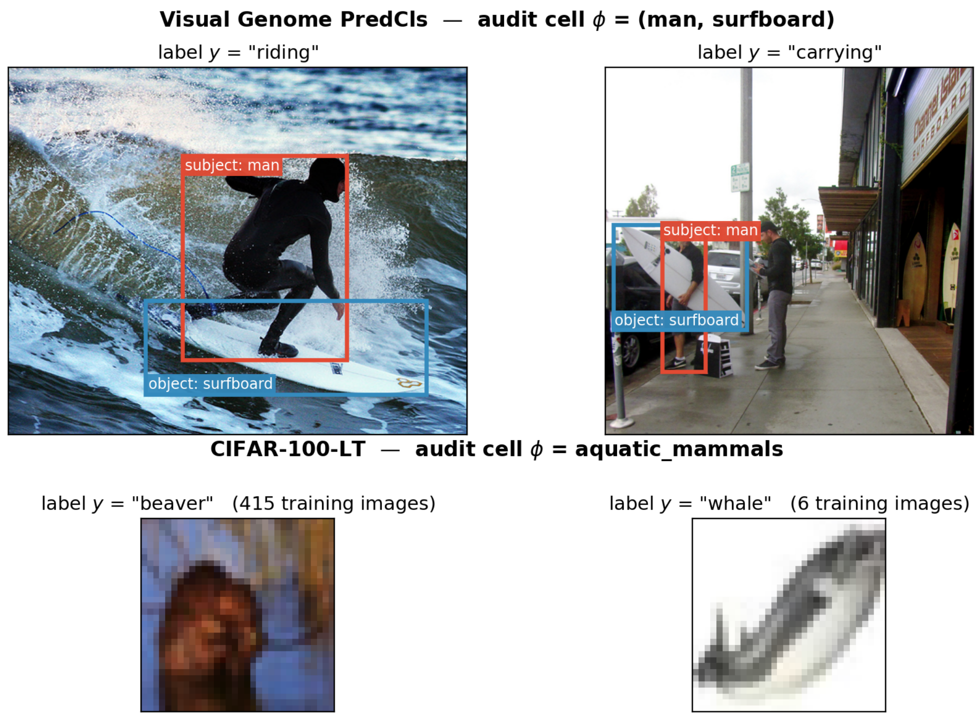

Fig. 4 What an audit cell contains, on both benchmarks. Each row shows two real test examples that share the same declared prior feature ϕ but carry different labels – precisely the configuration within-cell transport exploits. *Top:* two Visual Genome relations from the cell $\phi = (\mathrm{man}, \mathrm{surfboard})$. The class pair is identical and therefore cannot distinguish "riding" from "carrying"; only the image content can, which is what ΔR is designed to credit. Boxes are the annotated subject (red) and object (blue) regions that MLP-SPATIAL turns into geometry features and MLP-VISUAL crops and encodes with CLIP. *Bottom:* two CIFAR-100 test images from the superclass ϕ = aquatic_mammals, drawn from opposite ends of the long-tailed training distribution (415 versus 6 training images). Crossing the labels within a row preserves the cell and its label histogram while destroying which prediction belongs with which instance.

Five deliberately controlled predictors expose different information. FREQ is the training-count distribution conditioned on the class pair. FREQ+OVERLAP additionally uses a binary box-overlap indicator. MLP-CLASS consumes fixed subject/object class embeddings and is therefore a function of ϕ alone. MLP-SPATIAL adds 12 normalized box-geometry features (relative center, scale, aspect, intersection, containment, and IoU); MLP-SPATIAL+ increases its capacity. All models are trained once on the training split. WAGER consumes only their cached test distributions. We use the quadratic score, image-clustered intervals, and 499 within-cell cyclic randomizations.

4.3 Direct gain attribution

Table 2 and Figure 5 give the central result. MLP-CLASS improves the quadratic score over FREQ by 0.03386, but its gain vector is constant inside every class-pair

Table 1 Frozen-model top-1 accuracy on VG150 PredCls. Similar accuracies do not reveal whether the difference is prior-recoverable.

Model	Accuracy
FREQ	0.6564
FREQ+OVERLAP	0.6545
MLP-CLASS	0.6554
MLP-SPATIAL	0.6639
MLP-SPATIAL+	0.6674

Table 2 WAGER decomposition on 227,337 identified VG150 PredCls relations. Gains use the quadratic score. CI is image-cluster robust; p is the 499-shift relation-level randomization diagnostic. "–" denotes an undefined share because total gain is negative.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	$\widehat{\rho}$	p
FREQ+OVERLAP	FREQ	−0.01229	−0.01430	0.00202 $[0.00168, 0.00235]$	–	.002
MLP-CLASS	FREQ	0.03386	0.03386	0.00000 $[0.00000, 0.00000]$	0.00	.724
MLP-SPATIAL	FREQ	0.04270	0.02832	0.01439 $[0.01373, 0.01504]$	0.34	.002
MLP-SPATIAL+	FREQ	0.04644	0.02885	0.01760 $[0.01686, 0.01834]$	0.38	.002
MLP-SPATIAL	MLP-CLASS	0.00885	−0.00554	0.01439 $[0.01373, 0.01504]$	1.63	.002
MLP-SPATIAL+	MLP-SPATIAL	0.00374	0.00053	0.00321 $[0.00289, 0.00353]$	0.86	.002

cell. WAGER therefore assigns the entire improvement to prior transport and exactly zero to instance alignment. This is an algebraic consequence of the estimator, not a learned baseline passing a sanity check.

MLP-SPATIAL versus FREQ gains 0.04270 in total: 0.02832 is retained after label transport and 0.01439 is instance alignment (CI $[0.01373, 0.01504]$). The direct comparison MLP-SPATIAL versus MLP-CLASS is more revealing. Its total gain is only 0.00885, because prior fitting deteriorates by 0.00554, while spatial alignment improves by 0.01439. Thus the alignment share is 1.63: the useful instance-specific change is partially hidden by a worse prior component. A ratio constrained to $[0, 1]$ or thresholded at 0.5 would misdescribe this tradeoff.

MLP-SPATIAL+ adds 0.00374 over MLP-SPATIAL, of which 0.00321 is alignment (CI $[0.00289, 0.00353]$). FREQ+OVERLAP illustrates the opposite caution: overlap creates a small positive alignment term, but its negative prior term is larger, so the model is worse overall. WAGER does not label any use of image-derived information as an improvement; it reports both channels and their sign.

4.4 Sensitivity analyses

For MLP-SPATIAL versus MLP-CLASS, the conclusion survives every planned sensitivity check (Table 3). Changing from the quadratic to a probability-floored log score changes scale (0.01439 to 0.06536) but not sign or significance. Coarsening ϕ from the class pair to subject class increases the credited alignment to 0.02181. Proposition 1

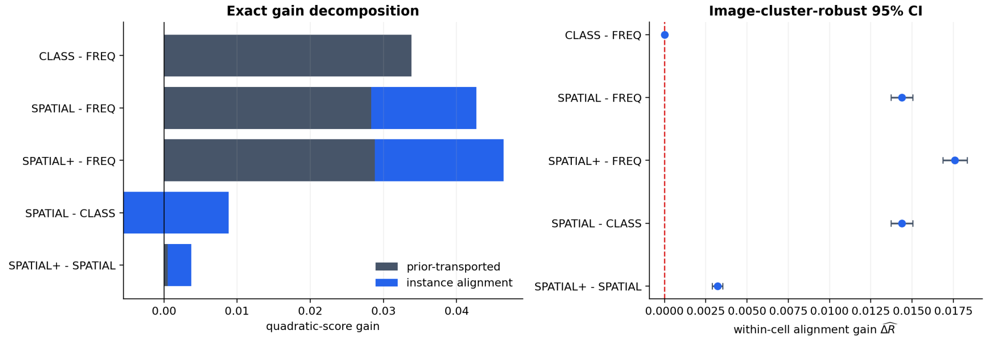

Fig. 5 **Direct model-pair attribution.** Left: exact decomposition of each positive gain into prior-transported and instance-alignment components (the negative prior term for SPATIAL–CLASS extends left of zero). Right: primary alignment estimates with image-cluster-robust intervals.

Table 3 Sensitivity of the MLP-SPATIAL vs. MLP-CLASS decomposition to score, audit granularity, and minimum cell size. CI is image-cluster robust on $\widehat{\Delta R}$.

Factor	Setting	$\widehat{\Delta T}$	$\widehat{\Delta R}$ (95% CI)	Cells
Score	Brier (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Score	Log (floored)	0.04297	0.06536 [0.06294, 0.06779]	6346
Audit cell	class pair (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Audit cell	subject only	0.00943	0.02181 [0.02079, 0.02283]	150
Min. cell size	2 (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Min. cell size	5	0.00772	0.01395 [0.01329, 0.01462]	4060
Min. cell size	10	0.00683	0.01330 [0.01262, 0.01398]	2744
Min. cell size	20	0.00570	0.01225 [0.01155, 0.01295]	1757

identifies why: the coarser cell's estimate equals the class-pair estimate's conditional average plus a between-class-pair covariance term, and that term is positive here, meaning some of the credited gain reflects class-pair-level structure that the finer audit had already correctly assigned to ΔP. This is precisely the sense in which the class-pair audit is the conservative choice: it is not that coarsening is wrong, but that it cannot be assumed neutral, and only the finer partition's estimate isolates alignment gain net of every prior regularity expressible in ϕ. Raising the minimum cell size from 2 to 5, 10, and 20 yields 0.01395, 0.01330, and 0.01225; all cluster-robust lower limits remain positive. Unlike the rejected projection-based formulation, there is no smoothing strength, split fraction, clipping constant, betting order, or gain-ratio decision threshold to tune.

4.5 Real-pixel signal on Visual Genome

Every model in Sections 4.2–4.4 is a function of the declared class pair and, for MLP-SPATIAL(+), of box geometry derived from the annotation, not of the image itself. Box geometry is a proxy for image content, and a sceptical reader is entitled to ask whether WAGER's alignment channel responds to genuine pixel evidence or only to that proxy. Figure 4 makes the question concrete: the cell $\phi = (\text{man}, \text{surfboard})$ contains both "riding" and "carrying", and while their box configurations do differ, what separates them is what the image shows.

We therefore add MLP-VISUAL, which crops the annotated subject and object regions from the original Visual Genome image and encodes each with a frozen CLIP ViT-B/32 image encoder [42]. The two 512-dimensional crop embeddings are concatenated with the same class embeddings used by MLP-CLASS, and only a small MLP head (256, 128 hidden units) is trained. CLIP is never fine-tuned, so any alignment gain reflects information CLIP's pretraining already extracted from pixels and the head merely matched to the SGG label, not representation learning on VG itself.

Matched training data.

This comparison is reported separately from Table 2 rather than as an extra row in it, for a reason that is itself an instance of the paper's argument. Encoding every training relation with CLIP is costly, so MLP-VISUAL is trained on a subsample; comparing it against the full-data MLP-SPATIAL would then confound feature content with training-set size, and a disappointing result could not be attributed to the pixels. We instead retrain all three compared predictors on one fixed, seeded subsample of 100,000 of the 532,987 training relations, denoting them MLP-CLASS-S, MLP-SPATIAL-S and MLP-VISUAL-S. They differ only in their feature set, so any difference between them isolates the features. The audit itself still uses the complete 229,605-relation test split, the same class-pair ϕ, and image-clustered intervals, so it remains directly interpretable alongside the main analysis. Appendix D gives the sampling procedure, the crop-deduplication scheme, and the count of images that could not be retrieved.

4.6 Cross-domain validation: long-tailed image classification

Sections 4.2–4.5 are all instances of one task (predicate classification on an ordered object pair). Section 2.2 argued that long-tailed image classification poses a structurally identical question: a class-frequency prior can be fit better without any per-instance improvement. We test this directly on CIFAR-100-LT [43], an exponential-imbalance (ratio 100) subsample of CIFAR-100's training split with the original, balanced test split retained for evaluation. We train ResNet-32 [44] classifiers that differ only in their per-class loss weighting: a vanilla cross-entropy baseline (CE); class-balanced effective-number re-weighting applied from initialization (CB) [43]; and the same re-weighting deferred to the final twenty epochs (DRW) [45]. Optimizer, schedule, augmentation, epoch budget, and seed are identical across runs (Appendix D).

Choosing ϕ.

The declared prior feature must predict the label without determining it. Taking ϕ to be the fine class label itself would make every audit cell label-homogeneous, so within-cell transport would be vacuous and $\widehat{\Delta R}$ would be identically zero as an algebraic artifact rather than as a finding. We therefore use CIFAR-100's own 20 coarse superclasses, each containing five fine labels – structurally the same situation as a VG class-pair cell containing several predicates (Figure 4). Each superclass cell holds 500 balanced test images. We additionally report the trivial coarsening to a single global cell, which audits against the global class marginal and instantiates Proposition 1 in a second domain, and a head/mid/tail training-frequency tier partition. Because CIFAR-100 test images are independent (unlike VG's multiple relations per image), no cluster grouping is needed and the plain Hájek interval of Section 3.7 applies.

Result.

Top-1 accuracies are 0.3662 (CE), 0.2627 (CB), and 0.3874 (DRW): re-weighting from initialization degrades the classifier badly, while the deferred schedule delivers the expected improvement. Table 4 decomposes both gains.

The CB run shows the confound in its starkest form. Its aggregate quadratic-score gain over CE is essentially nil ($\widehat{\Delta T} = +0.00143$), which in isolation would read as a harmless reparameterization. WAGER shows it is nothing of the sort: within superclasses the prior-transported channel gains +0.19713 while instance alignment *loses* −0.19569 (95% CI [−0.20728, −0.18411]), two large and nearly cancelling effects. Re-weighting bought a much better fit to the within-superclass class-frequency distribution and paid for it with per-instance discrimination, as the ten-point accuracy drop confirms. The near-zero total is a coincidence of two opposing channels, exactly the confound the method exists to expose; the score does not fall with accuracy because CE is badly overconfident (mean maximum probability 0.764 at 0.3662 accuracy), so CB's softer predictions recover on calibration what they lose on alignment.

DRW is the more representative case, because it is a real improvement of the kind a benchmark submission would report. Its gain of $\widehat{\Delta T} = +0.06606$ splits into $\widehat{\Delta P} = +0.04206$ and $\widehat{\Delta R} = +0.02400$ (95% CI [+0.01340, +0.03460], randomization $p = .002$). Roughly two thirds of the improvement is recoverable from the class-frequency prior alone, but the remaining third is a genuine, statistically significant gain in per-instance discrimination – deferred re-weighting does more than re-rank the classes. Neither the accuracy delta nor the aggregate score could have separated these.

The tier breakdown in Table 5 and Figure 6 localizes that alignment gain. For DRW it is concentrated where the long-tail literature intends it: +0.09009 on few-shot classes and +0.06760 on medium-shot, against −0.07624 on many-shot classes, which lose alignment as the schedule shifts capacity away from the head. CB shows the same ordering with every level shifted downward, its many-shot alignment collapsing to −0.41977. WAGER thus separates not only how much of a long-tail gain is genuine, but which part of the frequency spectrum it comes from.

This also supplies an external check on the theory. Computing the alignment term directly from the covariance identity of Eq. (7) – a code path that shares nothing with the transport estimator – gives −0.19530, and multiplying by the exact factor $n_c/(n_c - 1)$ of Proposition 2 at $n_c = 500$ gives −0.195693, matching the estimator's −0.19569. Theorem 1 and Proposition 2 are thus confirmed numerically on real trained models, not only on the synthetic populations of the unit tests.

Coarsening behaves as Proposition 1 predicts. Moving from the 20 superclass cells to a single global cell changes the credited alignment from −0.19569 to −0.25430; the difference is the between-cell covariance term, which is negative here, so the coarse audit overstates the alignment loss by attributing to it structure that the superclass

Table 4 WAGER decomposition on the 10,000-image balanced CIFAR-100-LT test split. Gains use the quadratic score; CI is the Hájek interval on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic. "–" marks an undefined share (non-positive total gain).

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (20)	0.00143	0.19713	−0.19569 [−0.20728, −0.18411]	1.000
CB vs CE	global (1)	0.00143	0.25573	−0.25430 [−0.26801, −0.24058]	1.000
CB vs CE	tier (3)	0.00143	0.25247	−0.25103 [−0.26453, −0.23754]	1.000
DRW vs CE	superclass (20)	0.06606	0.04206	0.02400 [0.01340, 0.03460]	.002
DRW vs CE	global (1)	0.06606	0.03522	0.03085 [0.01860, 0.04309]	.002
DRW vs CE	tier (3)	0.06606	0.03702	0.02904 [0.01702, 0.04107]	.002

Table 5 Alignment gain by training-frequency tier at the primary superclass audit. The deferred schedule concentrates its genuine per-instance improvement on rare classes; both methods lose alignment on the head.

		CB vs CE	DRW vs CE	
Tier	n	$\widehat{\Delta R}$	$\widehat{\Delta T}$	$\widehat{\Delta R}$
Few-shot (<20)	3000	−0.03337	0.13785	0.09009
Medium-shot (20–100)	3500	−0.11075	0.10799	0.06760
Many-shot (>100)	3500	−0.41977	−0.03740	−0.07624

partition correctly assigns to ΔP. The finer partition is again the conservative reading, for the same reason as in Section 4.4 and now in a second domain.

5 Discussion and Limitations

5.1 What the decomposition does and does not establish

WAGER changes the unit of analysis from a model to a model-pair gain. This matters scientifically. A class-only network may be a better prior estimator than FREQ, but that fact does not become evidence of instance-specific visual use. Conversely, a model can improve visual alignment while losing enough prior score to obscure the improvement in its aggregate metric. The signed identity $\Delta T = \Delta P + \Delta R$ exposes both cases without a separately fitted reference forecaster.

The interpretation remains conditional on the declared audit feature ϕ. A positive ΔR means the new model aligns its probability changes with the correct example better than the old model among examples sharing ϕ. It does not prove causal reasoning, compositional understanding, or robustness. If an unrecorded shortcut varies inside a cell, WAGER will count its predictive use in ΔR. Applications should therefore choose ϕ before looking at results, justify it from the benchmark's data generation process, and report finer and coarser alternatives. The observed increase under subject-only cells illustrates Proposition 1: coarsening adds a between-cell covariance term of either

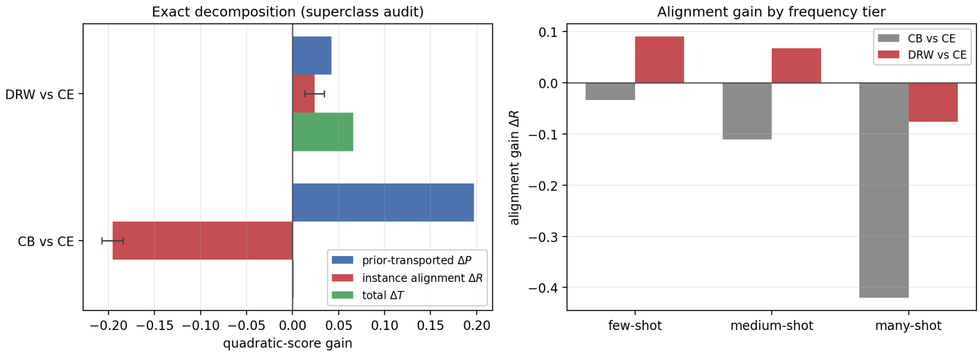

Fig. 6 **The same attribution transfers to a second task.** Left: exact decomposition at the superclass audit. The CB total gain (green) is visually almost absent at $+0.00143$ while its two channels reach ± 0.2 – a near-zero aggregate score concealing two large opposing effects. DRW's smaller total is genuinely split between both channels. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: the alignment channel by training-frequency tier, showing that DRW's genuine per-instance gain is concentrated on rare classes and negative on the head.

sign, so it cannot be assumed to leave the estimate unchanged, and only the finest identified partition isolates alignment gain net of every prior regularity expressible in ϕ. This is why the finest defensible prior feature is the conservative primary choice, not merely an empirical convention.

Exact transport also has a positivity requirement: a cell needs at least two examples. WAGER reports the identified fraction and makes no claim for singletons. Very fine ϕ can therefore improve confound control while reducing coverage. Approximate matching could extend the method to continuous priors, but would replace the exact identity's clear intervention with modeling assumptions; we leave that extension open.

The image-clustered confidence interval is asymptotic. The cyclic randomization test is finite-group exact only under its sharper within-cell exchangeability null and treats relations as the randomized units, so it is secondary in the SGG analysis. A future image-level randomization design that preserves several relation-cell margins simultaneously would strengthen finite-sample inference for structured benchmarks.

Finally, WAGER evaluates probability vectors, not only top-1 decisions. This is a feature—the method can detect useful redistribution below the argmax—but it requires released logits or probabilities. The quadratic score is bounded and strictly proper; other proper scores answer scale-dependent variants of the same question and should be declared in advance.

5.2 Broader applicability

The algorithm needs only four aligned arrays: two models' probability vectors, labels, and prior-cell identifiers, and nothing in Sections 3.1–3.7 is specific to relation prediction. Section 4.6 tests this directly outside SGG, on long-tailed image classification, taking the benchmark's own coarse superclasses as ϕ; the resulting decomposition separates a re-weighting schedule whose near-zero aggregate score hides two large cancelling channels from one whose genuine improvement is part prior-refitting and part rare-class alignment, which gives this claim empirical content rather than leaving it as an analogy. That case also illustrates the constraint on ϕ in a second domain: the fine class label cannot serve as the audit cell, because it would make every cell label-homogeneous and force ΔR to zero as an algebraic artifact rather than as a finding. For VQA, ϕ can similarly be a predeclared question template; for other multimodal benchmarks, it can encode answer format or prompt family – we do not evaluate these here, and flag them as the natural next domains rather than claim coverage we have not tested. In every case the scientific claim must remain precise: WAGER measures new prediction–label alignment within the declared cells. Its value is not that one choice of ϕ defines reasoning universally, but that every attribution becomes an explicit, reproducible counterfactual tied to a declared nuisance structure, and that this same construction produces a sensible answer on at least two structurally different tasks without modification.

6 Conclusion

We introduced WAGER, Within-cell Antisymmetric Gain Evaluation of Reasoning, a direct algorithm for attributing the score improvement between two vision models. By crossing labels among examples with the same prior feature, WAGER constructs the prior-only counterfactual from the evaluated data themselves and obtains an exact identity between total, prior-transported, and instance-alignment gains. The residual is an interpretable U-statistic, equals a conditional covariance gain under the quadratic score, and is computable in $O(NK)$ time. Controlled experiments support null calibration and signal recovery. On Visual Genome, the method assigns a class-only improvement entirely to the prior channel and identifies the added alignment produced by spatial geometry. Applied unchanged to long-tailed classification, it distinguishes a re-weighting schedule whose near-zero aggregate score hides two large cancelling channels from one whose genuine improvement is part prior-refitting and part rare-class alignment – a separation no aggregate score or accuracy delta can make. This gain-specific mechanism, rather than a combination of a frequency baseline and a generic significance test, is the paper's technical contribution.

Data Availability

All code and data used in this study are publicly available at https://github.com/vinhqdang/WAGER.

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Appendix A Proofs

A.1 Proof of Theorem 1

Condition throughout on $C = c$. Expanding over labels gives

$$\Delta T_c = \sum_y \mathbb{E}[H_X(y)\mathbb{1}\{Y = y\}], \qquad \Delta P_c = \sum_y \mathbb{E}[H_X(y)] \Pr(Y = y).$$

Subtracting yields the covariance sum in Eq. (6); the first identity is the definition of ΔR_c. Under the quadratic score,

$$H_X(y) = 2\Delta q_y(X) - \{\|q_1(\cdot \mid X)\|_2^2 - \|q_0(\cdot \mid X)\|_2^2\}.$$

The bracketed term is independent of the label index. Its contribution summed over y is $\mathrm{Cov}(B(X), \sum_y \mathbb{1}\{Y = y\}) = \mathrm{Cov}(B(X), 1) = 0$. The remaining terms give Eq. (7). If $H_X(y) = H_c(y)$ almost surely, every covariance is zero. $\square$

A.2 Proof of Theorem 2

Summing the kernel in Eq. (12) over unordered pairs, every observed term $H_i(y_i)$ occurs in $n_c - 1$ pairs with coefficient $1/2$, while the two crossed terms together enumerate every ordered pair $H_i(y_j)$, $i \neq j$, with coefficient $1/2$. Since $\binom{n_c}{2} = n_c(n_c-1)/2$,

$$\widehat{\Delta R}_c = \frac{1}{n_c}\sum_i H_i(y_i) - \frac{1}{n_c(n_c-1)}\sum_{i \neq j} H_i(y_j) = \widehat{\Delta T}_c - \widehat{\Delta P}_c.$$

This is deterministic. Under exchangeable sampling, the average of the symmetric order-two kernel is the standard unbiased U-statistic for its population expectation [38]. $\square$

A.3 Proof of Proposition 1

Fix y and condition throughout on $\bar{\phi} = \bar{c}$. The law of total covariance applied to the pair $(H_X(y), \mathbb{1}\{Y = y\})$ with the finer conditioning variable C nested inside $\bar{c}$ gives

$$\begin{aligned}\mathrm{Cov}\big(H_X(y), \mathbb{1}\{Y = y\} \mid \bar{\phi} = \bar{c}\big) &= \mathbb{E}\big[\mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C) \bigm| \bar{\phi} = \bar{c}\big] \\ &\quad + \mathrm{Cov}\big(\mathbb{E}[H_X(y) \mid C], \Pr(Y = y \mid C) \bigm| \bar{\phi} = \bar{c}\big),\end{aligned}$$

using $\mathbb{E}[\mathbb{1}\{Y = y\} \mid C] = \Pr(Y = y \mid C)$. Summing over $y = 1, \ldots, K$ and applying the population identity of Theorem 1 to the left side (at $\bar{\phi} = \bar{c}$) and to the first term on the right side (at $C = c$, then averaged over c within $\bar{c}$) yields Eq. (8). $\square$

A.4 Proof of Proposition 2

Fix a cell c with $n_c \geq 2$ and $i \in c$. By definition of the label count, $\sum_{y=1}^K m_{cy} H_i(y) = \sum_{j \in c} H_i(y_j) = H_i(y_i) + \sum_{j \neq i} H_i(y_j)$, and Eq. (14) gives $\sum_{j \neq i} H_i(y_j) = (n_c - 1)\hat{P}_i$. Hence

$$\tilde{P}_i = \frac{1}{n_c}\sum_y m_{cy} H_i(y) = \frac{H_i(y_i) + (n_c - 1)\hat{P}_i}{n_c}.$$

Substituting $H_i(y_i) = \hat{P}_i + \hat{R}_i$ gives $\tilde{P}_i = \hat{P}_i + n_c^{-1}\hat{R}_i$, and $\tilde{R}_i = H_i(y_i) - \tilde{P}_i = \hat{R}_i - n_c^{-1}\hat{R}_i = (n_c - 1)n_c^{-1}\hat{R}_i$. Averaging $\tilde{R}_i$ within cell c and reweighting by n_c across cells, as in the dataset-level definition of $\widetilde{\Delta Z}$, gives $\widetilde{\Delta R} = N_*^{-1}\sum_c n_c \cdot \frac{n_c - 1}{n_c}\widehat{\Delta R}_c = N_*^{-1}\sum_c (n_c - 1)\widehat{\Delta R}_c$. $\square$

Appendix B Efficient influence computation

The implementation does not instantiate all pairs. In addition to Eq. (14), let $G_c(y) = \sum_{j\in c} H_j(y)$ and $O_c = \sum_{j\in c} H_j(y_j)$. The pair-kernel mean for observation i is

$$\bar{A}_i = \frac{1}{2}\left[H_i(y_i) + \frac{O_c - H_i(y_i)}{n_c - 1} - P_i - \frac{G_c(y_i) - H_i(y_i)}{n_c - 1}\right].$$

Therefore both the point estimate and Hájek projection require only label counts, column sums, and matrix–vector products inside each cell. For image clusters g, the implemented variance is

$$\widehat{\mathrm{Var}}(\widehat{\Delta R}) = \frac{G}{G-1}\frac{1}{N_*^2}\sum_{g=1}^{G}\left(\sum_{i\in g}\widehat{\psi}_i\right)^2,$$

where G is the number of images and $\widehat{\psi}_i$ is Eq. (16). We use a normal critical value; with 27,955 image clusters, small-G corrections are immaterial.

Appendix C Randomization algorithm

For each cell, sort examples once and index them $0, \dots, n_c - 1$. Randomization b draws $o_c^{(b)}$ uniformly from this index set and maps label position r to $(r + o_c^{(b)}) \bmod n_c$. Independent offsets across cells define an element of the product of cyclic groups. The cell label histogram is unchanged, so $\sum_y m_{cy} H_i(y)$ in Eq. (14) is precomputed; only the assigned label lookup changes. One randomization is $O(N)$ after the $O(NK)$ setup. The plus-one Monte Carlo correction includes the observed assignment and prevents zero p-values.

Appendix D Implementation and reproducibility

The NumPy implementation is in `wager/antisymmetric.py`. The primary routine `decompose_gain` validates probability matrices, forms score contrasts, excludes singleton cells, computes all three components, verifies the identity numerically, and returns per-instance transported/alignment values plus intervals. The Visual Genome driver is `experiments/run_sgg_wager.py`; controlled validation and sensitivity checks are in `experiments/antisymmetric_simulation.py` and `experiments/antisymmetric_ablations.py`. All random seeds are fixed. Unit tests cover the decomposition identity, quadratic covariance identity, cell-constant null, singleton reporting, cluster inference, randomization power, and log-score path, plus direct numerical checks of Propositions 2 and 1 (the latter against an explicit discrete population with exact expectations, independent of the estimator's own code path).

D.1 Compute environment for Sections 4.5–4.6

Model training and feature extraction for both new experiments ran on a single NVIDIA T4 (15 GB) GPU. The WAGER decomposition itself is pure NumPy and requires no accelerator: it consumes the resulting cached probability arrays, as in Section 4.2. All Python code runs under a pinned environment (Python 3.13).

Matched-subsample models (Section 4.5).

The three compared predictors share a fixed, seeded (seed = 0) subsample of 100,000 of the 532,987 training relations, drawn without replacement uniformly at the relation level (not the image level); the full 229,605-relation test split is always used for evaluation. All three use the identical head and optimizer – AdamW, learning rate 10^{-3}, weight decay 10^{-5}, 15 epochs, batch size 4096, hidden sizes $(256, 128)$, on standardized features – so they differ only in their input features.

Subject/object crops are taken from the annotated boxes and CLIP-preprocessed (bicubic resize to 224×224 with CLIP's published normalization constants); a crop with no valid overlap with the image falls back to a blank crop rather than being dropped, so every relation contributes a feature row. CLIP runs in half precision with a batch size of 512 crops and is never fine-tuned. Two implementation details are worth recording. First, the pretrained OpenAI weights were trained with the QuickGELU activation, so the encoder must be instantiated as `ViT-B-32-quickgelu`; loading the same weights under the plain `ViT-B-32` configuration silently substitutes a standard GELU and degrades every embedding. Second, because an annotated object box participates in several relations, crops are deduplicated by (image, box) before encoding and gathered afterwards, which removes roughly a third of the encoder work without changing any feature. Only the images referenced by the audited relations are retrieved, individually rather than through the two bulk archives.

ResNet-32 pair (Section 4.6).

CIFAR-100-LT uses the standard exponential-profile construction of Cui et al. [43] at imbalance ratio 100: class c (zero-indexed, $0, \ldots, 99$) keeps $n_c = \lfloor 500 \cdot \mu^c \rfloor$ training images, where $\mu = 100^{-1/99}$, so class 0 keeps all 500 images and class 99 keeps 5; the balanced 10,000-image test split is untouched. Both models are a from-scratch, randomly initialized CIFAR ResNet-32 (three stages of five basic residual blocks, 16/32/64 channels) trained for 80 epochs with SGD (momentum 0.9, Nesterov, weight decay 2×10^{-4}), batch size 128, standard crop-and-flip augmentation, and a cosine learning-rate schedule (peak 0.1, five-epoch linear warmup). The only difference between the two runs is the per-example loss weight: uniform for the baseline, and the effective-number class-balanced weight $w_c \propto (1 - \beta)/(1 - \beta^{n_c})$ with $\beta = 0.9999$ for the class-balanced model, renormalized to average to one across classes.